Academic Year 2026-27
FY MCA- SEM I-DIV A
IT12 Data Structure and Algorithms
Practical Assignment2

Roll No: 2601072 **Name: Ashwini Amit Gund.**
Date: 2/11/2026

Q1.

```
# Stack Implementation.
#1.push 2.Pop 3.Display 4.Reverse string using stack 5.Reverse stack using stack

stack = []

while True:
    print("\n----- STACK OPERATIONS -----")
    print("1. Push")
    print("2. Pop")
    print("3. Display")
    print("4. Reverse String using Stack")
    print("5. Reverse Stack using Another Stack")
    print("6. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        element = int(input("Enter element to push: "))
        stack.append(element)
        print("Element pushed successfully!")

    elif choice == 2:
        if len(stack) == 0:
            print("Stack is Empty!")
        else:
            print("Popped Element:", stack.pop())

    elif choice == 3:
        if len(stack) == 0:
            print("Stack is Empty!")
        else:
            print("Stack Elements:", stack[::-1])

    elif choice == 4:
```

```
            string = input("Enter a string: ")
        temp = []

        for ch in string:
            temp.append(ch)

        reverse = ""
        while temp:
            reverse += temp.pop()

        print("Reversed String:", reverse)

    elif choice == 5:
        if len(stack) == 0:
            print("Stack is Empty!")
        else:
            temp = []

            while stack:
                temp.append(stack.pop())

            while temp:
                stack.append(temp.pop())

            print("Stack Reversed Successfully!")
            print("Reversed Stack:", stack[::-1])

    elif choice == 6:
        print("Exiting Program...")
        break

    else:
        print("Invalid Choice!")
```

Q2.

Linear Queue Implementation

#1.Enqueue 2.Dequeue 3.Display 4.Reverse queue using stack 5.Merge 2 queues 6.Check if two queues are identical.

```
while True:
    print("\n----- LINEAR QUEUE OPERATIONS -----")
    print("1. Enqueue")
    print("2. Dequeue")
    print("3. Display")
    print("4. Reverse Queue using Stack")
    print("5. Merge Two Queues")
    print("6. Check Two Queues are Identical")
    print("7. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        element = int(input("Enter element: "))
        queue.append(element)
        print("Element inserted successfully!")

    elif choice == 2:
        if len(queue) == 0:
            print("Queue is Empty!")
        else:
            print("Deleted Element:", queue.pop(0))

    elif choice == 3:
        if len(queue) == 0:
            print("Queue is Empty!")
        else:
            print("Queue Elements:", queue)

    elif choice == 4:
        if len(queue) == 0:
            print("Queue is Empty!")
        else:
            stack = []
```

```
                while queue:
            stack.append(queue.pop(0))

        while stack:
            queue.append(stack.pop())

        print("Queue Reversed Successfully!")
        print("Reversed Queue:", queue)

elif choice == 5:
    queue2 = []
    n = int(input("Enter number of elements in second queue: "))

    for i in range(n):
        element = int(input("Enter element: "))
        queue2.append(element)

    queue.extend(queue2)

    print("Merged Queue:", queue)

elif choice == 6:
    queue2 = []
    n = int(input("Enter number of elements in second queue: "))

    for i in range(n):
        element = int(input("Enter element: "))
        queue2.append(element)

    if queue == queue2:
        print("Both Queues are Identical!")
    else:
        print("Both Queues are Not Identical!")

elif choice == 7:
    print("Exiting Program...")
    break

else:
    print("Invalid Choice!")
```